

Multiple Choice (10 marks)

1. The emission spectrum of the doubly ionized lithium atom Li^{++} ($Z = 3$, $A = 7$) is identical to that of a hydrogen atom in which all the wavelengths are
 - (a) decreased by a factor of 9
 - (b) decreased by a factor of 49
 - (c) decreased by a factor of 81
 - (d) increased by a factor of 9
2. The rest mass of a particle with total energy 5.0 GeV and momentum 4.9 GeV/c is approximately
 - (a) 0.1 GeV/ c^2
 - (b) 0.2 GeV/ c^2
 - (c) 0.5 GeV/ c^2
 - (d) 1.0 GeV/ c^2
3. Which of the following lasers utilizes transitions that involve the energy levels of free atoms?
 - (a) Diode laser
 - (b) Dye laser
 - (c) Free-electron laser
 - (d) Gas laser
4. For which of the following thermodynamic processes is the increase in the internal energy of an ideal gas equal to the heat added to the gas?
 - (a) Constant temperature
 - (b) Constant volume
 - (c) Constant pressure
 - (d) Adiabatic
5. Which of the following gives the total spin quantum number of the electrons in the ground state of neutral nitrogen ($Z = 7$)?
 - (a) 1/2
 - (b) 1
 - (c) 3/2
 - (d) 5/2

True / False (10 marks)

Answer True or False

6. True or False: If the total energy of a particle is twice its rest energy, its relativistic momentum is equal to $(\sqrt{3})mc$.
7. True or False: Electrical resistivity measurements can directly indicate the type of charge carriers in a semiconductor.
8. True or False: In a constant pressure process, the increase in the internal energy of an ideal gas is equal to the heat added to the gas.
9. True or False: A meter stick moving at $0.8c$ takes 1.6 ns to pass an observer in their reference frame.
10. True or False: During a reversible thermodynamic process, the internal energy of the system always increases.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the work required to accelerate a proton from rest to a speed of $0.6c$. Explain the underlying physics principles and derive the expression for the work done in terms of the proton's mass.
12. Discuss the measurements an astronomer can make when observing a moon orbiting a planet, and explain why the mass of the moon cannot be determined from these measurements.

End of Paper